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Post-Mortem Human Identification Using Chest X-Ray and CT Scan Images

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Abstract: *In this paper, we develop a computerized human post-mortem identification system by using chest biometrics feature. The main purpose is to identify an unknown person after death. An unknown death body is identified by comparing the chest CT image after death with the X-ray images before death stored in a database to find the highest similarity. The proposed system consists of four main processes: pre-processing, boundary extraction, feature extraction, and similarity calculation and ranking results. All the images are firstly enhanced using Contrast Limited Adaptive Histogram Equalization (CLAHE), and then ribs boundaries are extracted using morphological erosion. After that, the features are extracted using Discrete Fourier Transform (DFT). We use the Euclidean distance to calculate similarity between those features, and then ranking is performed based on the resulted distances. Finally, the system retrieves the ante-mortem X-ray images that are similar to the query image of post-mortem CT image of a death person. Experiments are conducted on dataset collected from the Faculty of Medicine, University of Miyazaki, and our experimental results are compared with the best result of the existing system under the same conditions. From the comparison, our proposed system performs best and gives the accuracy of 74.07%.*
Keywords: *Human identification, Chest biometrics feature, Ante-mortem, Post-mortem, CLAHE, Boundary extraction, DFT, Similarity calculation, Ranking and Euclidean distance*

1. Introduction

From last decades, most of human identification and authentication systems have been made use of traditional authentication techniques such as passwords. Since it demands highly reliable person authentication systems, the field of biometrics has become an active and attractive research topic [1]. Moreover, biometrics has now gained a widespread acceptance to provide legitimate identity of an individual person, particularly computer-vision based applications such as surveillance, security, and forensics. Therefore, biometric features have been utilized by law enforcement agencies or criminal justice agencies for identification processes. Actually, biometrics is a term that uses some characteristics of human body such as fingerprint pattern, face pattern, retina, hand geometry, iris pattern, gait cycle, and voice pattern, for recognizing and identifying humans [2]. The biometric identification and verification methods can be divided into two categories: behavioral methods (such as speech, action, signature, and keyboard typing) and physiological methods (such as fingerprint, iris, DNA, and facial features) [3]. General identifications prior to death (known as ante-mortem) are possible through comparison of such behavioral biometrics (such as speech, and action). However, the identification of a person after death (called post-mortem identification) is impossible using some behavioral biometrics.

While behavioral biometrics are not suitable for Post-Mortem (PM) identification, most of the physiological biometrics are also not appropriate for PM identifi-

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cation. Mostly, human identification systems have been developed by utilizing conventional physiological biometrics such as face, fingerprint, and DNA appraisal. In 1985, the crash of Japan Airlines 123 flight happened. In that case, identifications of dental findings, fingerprints, and DNA approval become difficult because of the delay of finding corruptions and damages. Another is the case of the Great East Japan Earthquake (also known as the 2011 Tōhoku earthquake). The Japanese National Police Agency report confirmed 15,894 deaths. Among them, only 15,749 persons had been identified [4].

In such severe natural disaster circumstances, or sometimes when identification may take more than a couple of weeks due to the large number of victims, most of those physiological biometrics cannot be used for identification because of the decay of some soft tissues. To make post-mortem identification, a biometric identifier has to survive even under severe conditions. In our body, the most robust and corruption resistance part is the bone that can be obtained with X-ray or CT images at the time of medical checkup. Thus, in recent years, researchers have made post-mortem identification by exploiting radiographic images.

A lot of researches have been done for automatically identifying a death person based on dental biometrics [1,5,6], and lateral patella [7]. Among the victims, there may have people who had never been a dental treatment. In such a situation, it is difficult to confirm one's identity. Recently, some researchers have thus proposed image technology based models using a new biometrics identifier (chest biometrics) for human identification [4]. In [4], the authors proposed a stochastic approach for post-mortem human identification by utilizing chest X-ray and CT images. In that approach, the features were directly extracted from each original radiograph using Histogram of Gradients (HoG), Markov Feature Vector (MFV), and Bag of Words (BoW) methods. Then, the feature matching was performed based on Euclidean distance method. The system retrieved the top matched 15 images as output, and gave the accuracy of 63%.

From the literature, it has been noted that any luminance contrast information of an image was not adjusted even though radiographic images used in the system are poor in contrast and blurred. In addition, they used HoG as a part of features. Navneet Dalal and Bill Triggs [8] presented HoG descriptors in 2005. They mainly designed it for human pedestrian detection in static images. The HoG descriptor is thus particularly suited for human detection in images. Therefore, some features used in that system may be not appropriate. All these conditions lead to performance degradation.

Thus, our work mainly aims to improve the accuracy of post-mortem human identification system by producing an original contribution in the following areas: (a) pre-processing to handle low contrast problem; (b) boundary extraction to get more sharped ribs contour; and (c) feature extraction. Finally, we compare the results of our proposed system with those of the system proposed in [4].

The remaining sections of this paper are organized as follows. Section 2 explains detailed workflow of our proposed system and Section 3 describes our experimental environment and analysis results and system accuracy are discussed in Section 4. In Section 5, we conclude our approach for human identification and present some improvements as future work.

2. Proposed System and Methods

We propose a new approach for identifying the unknown individual's identity of a death person. The proposed system mainly consists of four steps:

1) Pre-processing
2) Boundary extraction
3) Feature extraction, and
4) Similarity calculation and ranking

Figure 1 shows the overview of the proposed system.

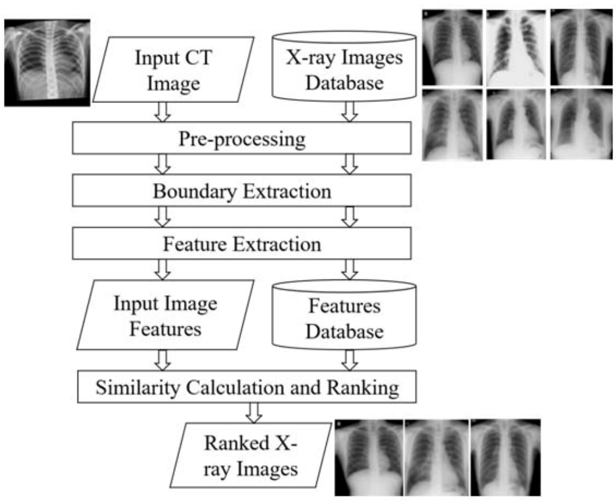


Figure 1. Overview of the proposed system

2.1. Pre-processing

Pre-processing is the first step in image processing and this step is very important because it is concerned with the quality of an image [1]. Since our radiographic images are poor in contrast, they are needed to be enhanced. Thus, the contrast of those images is firstly enhanced using the Contrast-Limited Adaptive Histogram Equalization (CLAHE). This method generally increases the local contrast of an image. The CLAHE procedure of image size $M \times N$ consists of the following steps [3].

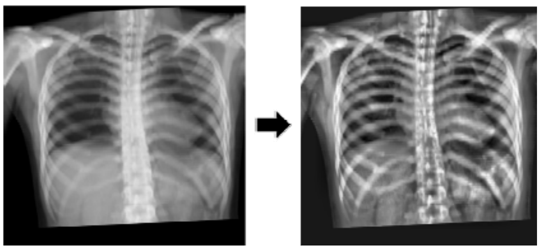
1) The image is divided into several non-overlapping regions of almost equal sizes.
2) The histogram of each region is calculated.
3) Based on a desired limit for contrast expansion, a clip limit for clipping histograms is obtained.
4) Each histogram is redistributed in such a way that its height does not go beyond the clip limit. The clip limit β is obtained by:

$$\beta = \frac{MN}{L}\left(1 + \frac{\alpha}{100}(s_{\max} - 1)\right) \tag{1}$$

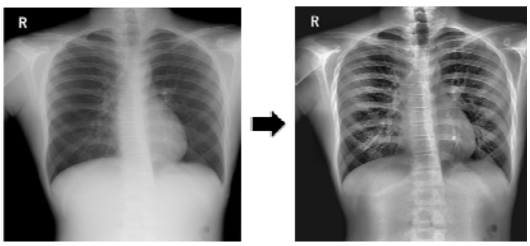
where α is a clip factor, $s_{\max}$ is maximum allowable slope, and L is the number of gray levels.

5) The Cumulative Distribution Functions (CDF) of the resultant contrast limited histograms are determined for grayscale mapping.

We improved the quality of images by setting the clip limit to 0.025 and using the 'Rayleigh Distribution' on CLAHE. Figure 2 shows the results of CLAHE applied on both CT and X-ray images of the same person (12th person) and the results of further processes will also be shown using that person.



(a) Original CT image and its enhanced image



(b) Original X-ray image and its enhanced image

Figure 2. Pre-processing results

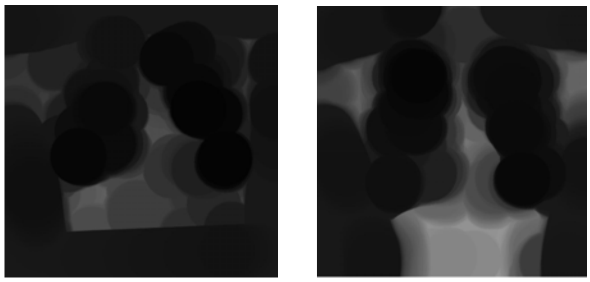
2.2. Boundary extraction

The next step is the boundary (contour) extraction of images. This step is done for the purpose of getting the more sharped ribs contour in the image. To detect boundaries in the image, the following equation [9] is used.

$$\beta(A) = A - (A \ominus B) \tag{2}$$

where $\beta(A)$ is a boundary extracted image, A is an image, B is a structuring element, and $(A \ominus B)$ is erosion of A by B.

According to the equation, we firstly applied morphological erosion to both the resulted enhanced CT and X-ray images. Then, we extracted ribs contour by subtracting the eroded image from its enhanced image. In this system, we used the structure element of 'disk shape' with radius = 50. The results of morphological erosion are shown in Figure 3 and boundary extraction results are shown in Figure 4.



(a) Eroded CT image (b) Eroded X-ray image

Figure 3. Morphological erosion results

(a) Boundary extracted CT image

(b) Boundary extracted X-ray image

Figure 4. Boundary extraction results

2.3. Feature extraction

After all the images are pre-processed, we extracted the features of each image. The feature extraction phase is the main process in our proposed system. In this system, we extracted the features of an image using one of the frequency domain techniques, the two-dimensional Discrete Fourier Transform (DFT) technique.

DFT is an efficient tool applied on images to separating an image into its sine waves and cosine waves. It is the output of the process of converting images from spatial domain into frequency domain. Normally image details are concentrated in low frequencies, and noise and edges are concentrated in high frequencies. In frequency domain image, we can distinguish the frequencies that represent the image; in addition we can know the frequencies that carry the noise [2].

Transformation or frequency domain techniques are based on the manipulation of the orthogonal transform of the image rather than the image itself. The orthogonal transform of the image has two components: magnitude and phase. The magnitude consists of the frequency content of the image. The phase is used to restore the image back to the spatial domain [10]. Therefore, the amplitude and phase features are extracted from each image in our system, and the results are shown in Figure 5. Then, the resulted two features are stored as feature vectors to be used for similarity matching.

For an image of size $M \times N$, the two-dimensional DFT is given by the equation [11]:

$$F(u,v) = \sum_{x=0}^{M} \sum_{y=0}^{N} f(x,y) e^{-i2\pi\left(\frac{ux}{M} + \frac{vy}{N}\right)} \tag{3}$$

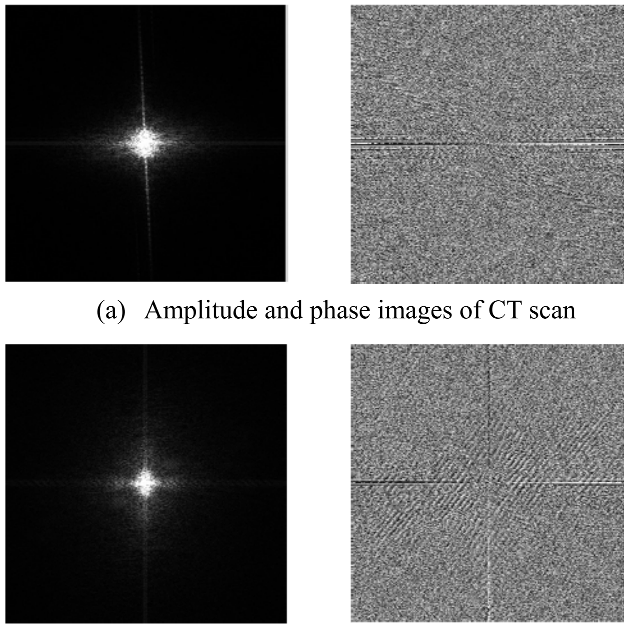
The amplitude (magnitude) feature of an image can be obtained by:

$$|F(u,v)| = \left[R(u,v)^2 + I(u,v)^2 \right]^{1/2} \tag{4}$$

The phase angle feature is given by:

$$\varphi(u,v) = \arctan\left(\frac{I(u,v)}{R(u,v)} \right) \tag{5}$$

where $f(x,y)$ is the input image in spatial domain, $F(u,v)$ is the output image in frequency domain, x, y are the input indexes of spatial domain, u, v are the input indexes of frequency domain, M, N represent the image dimensions, and $R(u,v)$ and $I(u,v)$ represent the real and imaginary components of $F(u,v)$.



(a) Amplitude and phase images of CT scan

(b) Amplitude and phase images of X-ray

Figure 5. Feature extraction results

2.4. Similarity calculation and ranking

Once the features are extracted from the images, it is needed to find the similarity between two images by using those features. There are many similarity measurement methods. In our system, we calculated the similarity measures between the feature vector of the query image and the features vector of images stored in the database using Euclidean distance method. The Euclidean distance, denoted by $D(x,y)$, is evaluated using the following equation:

$$D(x,y) = \sqrt{\sum_{i=1}^{n} \left(x_i - y_i \right)^2} \tag{6}$$

where x is the feature vectors of the CT image, y is the feature vectors of the X-ray images in database, and n is the dimension of feature vectors.

Finally, we performed ranking by sorting the resulted distance scores in ascending order. Then, the system retrieves the X-ray images that are similar to the query of CT scan image based on ranked distance scores.

3. Experiment Environment

In order to assess the performance of our proposed approach, we used the same dataset as used by the previous researcher [4]. The data are collected from the Faculty of Medicine, University of Miyazaki. The proposed method is applied to 27 people (subjects). All the images are in DICOM (Digital Imaging and Communications in Medicine) format. The chest X-ray radiographs are of size 2470 × 2470, and the CT scan images are 256 × 256 in resolution. Thus, before any processing, all the images are firstly converted into .JPG format, and resized into 512 × 512 pixels. Figure 6 and Figure 7 show the CT scan images for query and X-ray images in the database of 27 subjects that the system used for experiments.

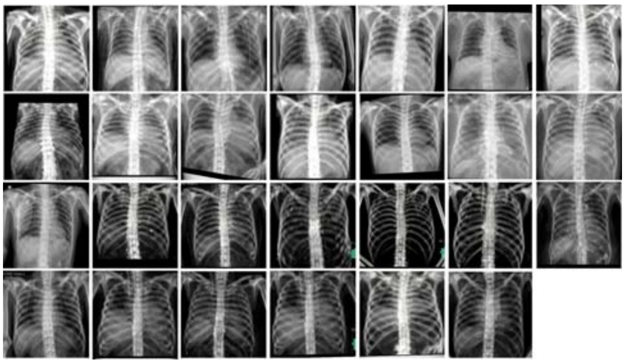


Figure 6. CT images after death

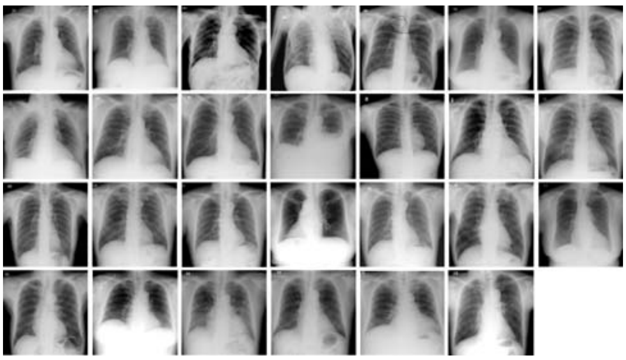


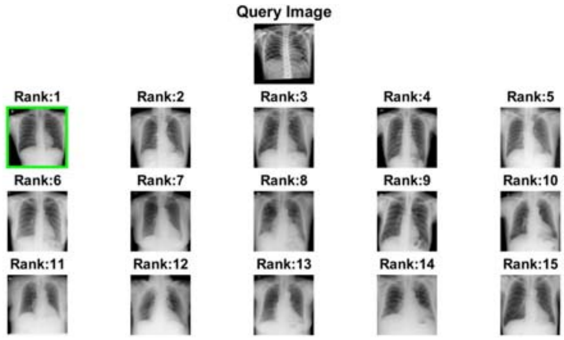
Figure 7. X-ray images before death

4. Results and Discussion

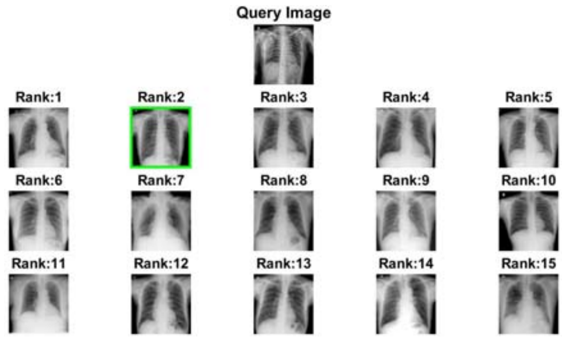
As a result, the proposed system produces and displays the top matched 15 people to the query image as output in order. In the system, there may be three kinds of retrieval results.

1) The matched image is in the top one (rank 1) position (best case) and its results are shown in Figure 8(a).
2) The matched image is contained within the top 15 positions shown in Figure 8(b).
3) The matched image is not contained in the top 15 positions (worst case) that is shown in Figure 8(c).

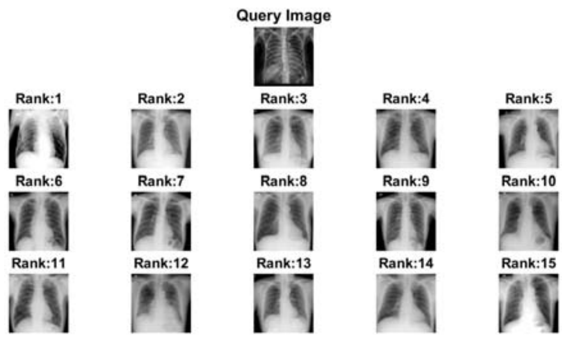
The first two cases are the cases that the system can accurately retrieve the X-ray images that match to the query image of CT image after death. In Figure 6, the images in the first two rows are low in contrast, but not too much, and the ones in last two rows are both very low in contrast and blurred. The query of Figure 8(a) is a little low in contrast and can see the ribs bone clearly even in the original image. After enhancing it, the ribs can be seen more clearly and it thus gives the matched image in rank 1. In case of the query image in Figure 8(c), it is too dark and blurred. Even though its contrast gets high after enhancing, it is still blurred. Thus, the matched image cannot be found within top 15. The system can show the matched image of not too much low contrast and not too blurred images within the top 15 images as output. Such kind of image results is like Figure 8(b). From the results of analysis, we can conclude that our proposed method depends on not only contrast but also blurring factors.



(a) Result of 12th person



(b) Result of 15th person



(c) Result of 21st person

Figure 8. Retrieval results of the system

The proposed system can correctly retrieve and identify 20 people among 27 people and this can be seen in Figure 9. It demonstrates the comparative ranking position of the proposed and existing systems. Then, accuracy of the system is computed dividing the number of correct matched people retrieved by the total number of people in the database. Table 1 illustrates the average retrieval rate from the proposed approach for comparative evaluation. From the results of Figure 9, we can see that our proposed system is able to retrieve for low contrast image (such as 25^{th} person, and 26^{th} person) that the existing system cannot.

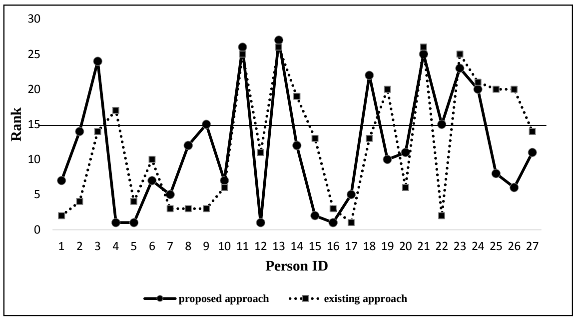


Figure 9. Comparative rank position of 27 people

Table 1. Comparison of experimental accuracy

Method	Accuracy
[Stedy MFV, BoW, Euclidean]	63%
[CLAHE, DFT, Euclidean]	74.07%

5. Conclusion

In this paper, we had developed the computerized post-mortem human identification system based on the chest biometrics feature. We had also proposed a new approach for identifying the unknown individual's identity of a deceased person. Moreover, we tested the proposed approach by comparing with the best result of previous researcher under the same environment. From the results of analysis, it can be seen that our proposed system gives more accuracy than the previous one.

Finally, our proposed system will be useful of identifying of a person who died because of various conditions such as natural disasters, fires, earthquakes, and airplane crashes, where conventional biometrics feature cannot be available to use. In the future, we will solve blurring problem and also try to conduct experiments on large-scale dataset of after and before death chest radiographs. Moreover, we will also explore new approaches or methods for pre-processing or feature extraction, which can lead to an improvement in retrieval accuracy. We hope our proposed system can give many benefits to our society.

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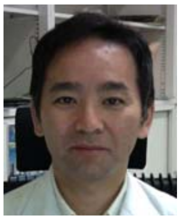

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